Power envelopes, exact defects, and cycle restrictions for the Juggler map

Publication draft. Not submitted.

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28 August 2026

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Abstract

The Juggler map is the nonlinear integer map

$$J(n) = \begin{cases} \lfloor \sqrt{n} \rfloor, & n \text{ even,} \\ \lfloor n^{3/2} \rfloor, & n \text{ odd.} \end{cases}$$

It is conjectured that every positive integer eventually reaches 1. We do not prove that conjecture.

We develop an exact finite-word calculus for the Juggler map. For every parity word w realized at n, the endpoint satisfies the power envelope

$$J^{|w|}(n)^{2^{|w|}} \le n^{3^{\#O(w)}}.$$

Hence $3^{\#O(w)} < 2^{|w|}$ forces $J^{|w|}(n) < n$ whenever $n \geq 2$. Local floor remainders lift through later exponents to an exact global defect. We prove its identity, vanishing classification, and composition law: concatenation is a two-term power-gap, and zero defect gives precisely the rigid monochrome towers. Exact inverse cells then yield necessary cycle restrictions, including superquadratic prefixes from a cycle minimum to later even states, and a small-cycle census: the Juggler map has no nontrivial cycle of length at most six. Length seven remains open.

As a secondary application, every even start and every odd start whose first image is even has a uniform one- or two-step descent certificate. A classical discrepancy estimate

$$\left|S_O(N)\right| \ll N^{5/6}, \qquad S_O(N) = \sum_{\substack{n \leq N \\ n \text{ odd}}} (-1)^{\lfloor n^{3/2} \rfloor},$$

implies that this uniform short-certificate class has density 3/4. An exact-linearization calculus for the nested floor powers extends the discrepancy method down the orbit: every itinerary word class of depth at most four except $OOO*$ has cardinality $2^{-|w|}N + O(N^{1-\delta_w})$ with explicit exponents, and consequently the class of starts carrying a uniform descent certificate of length at most four has natural density 13/16. An unconditional counting argument shows that parity equidistribution at all depths would give the set of starts with some finite descent certificate density one; the precise remaining obstacle at depth four is an explicit exponential-sum kernel, stated here as an open problem. None of these densities is a density of starts that reach 1.

All finite-word statements are conditional on the realized itinerary. The remaining pointwise question is whether almost every odd-to-odd start has some finite descent; no such theorem is proved here.

1. Introduction

The Juggler sequence was introduced by Pickover [1]; see also OEIS A094683 [2]. Universal arrival at 1 remains open.

The map combines a contracting even branch with an expanding odd branch,

$$E(n) = \lfloor n^{1/2} \rfloor, \qquad O(n) = \lfloor n^{3/2} \rfloor.$$

A word of length k with o odd letters has ideal exponent $3^o/2^k$. Floors are applied after every letter, and a word is available only when the orbit realizes those parities. The paper records the exact finite-word comparison, the compositional slack, the uniform short certificates, and the densities of the classes those certificates cover, through itinerary depth four.

The main contribution is the exact power-envelope and defect calculus, together with its inverse-cell and cycle consequences, chief among them a small-cycle census: no nontrivial cycle has length at most six. The certified-descent densities — 3/4 at two steps, 13/16 at four — are secondary corollaries. We do not prove a Collatz theorem or transfer Collatz stopping-time results to J.

1.1 Verification convention

Theorems in Sections 2–4 are proved both below and in Lean under the names listed at the start of each section. The analytic estimates of Section 5 (Theorems 5.1, 5.4, 5.7, 5.8, Proposition 5.5) and Proposition 6.1 are ordinary human proofs and are not formalized; the exact floor reductions beneath them — the parity bridge $\lfloor x \rfloor$ odd iff $\{x/2\} \geq 1/2$, and the gap-cell identity —

are Lean-verified (`floor_odd_iff_half_le_fract_half`, `floor_gap_eq_carry`, `seq_floor_gap` in `GapCells.lean`). The exact-linearization lemmas of Section 5 are additionally validated by scaled-integer computations recorded in the repository; those validations are checks, not proofs. Proposition 4.4 is an exact Python-integer census, not a theorem about an infinite set. The four-block chain named in Section 6 is Lean-certified (`four_block_pe_1999`); the failed reductions there are exact-integer counterexamples, not estimates. No finite computation is used as a termination proof.

1.2 Related maps

The nearest published comparison is the Collatz problem. Lagarias [3,4] surveys parity words and almost-all statements short of totality. Terras [5] and Everett [6] proved that almost every positive integer has a finite Collatz stopping time; Tao [7] later showed that almost all Collatz orbits attain almost bounded values. Those results are methodological cousins, not theorems about J. The Juggler analogue of Terras would be an almost-all descent theorem on the odd-to-odd class.

Crandall [8] and Matthews–Watts [9] treat piecewise-affine Hasse–Syracuse maps; the Juggler branches are floor powers, not affine. Prasad and Prasad [12] estimate excursion and stopping constants for juggler-like random walks; Section 6 keeps that comparison descriptive.

The discrepancy calculus of Section 5 concerns nested floor powers $\lfloor \lfloor n^c \rfloor^d \rfloor$. A literature check (August 2026) found no equidistribution result for such nested sequences: the Piatetski–Shapiro corpus covers single floors, intersections, and pseudo-polynomials. The tools themselves are classical — Vaaler's trigonometric approximation to the sawtooth [13] and van der Corput's second-derivative test in the form presented by Graham and Kolesnik [14].

2. Finite words, envelope, and defect

Let $\mathcal{B} = \{E, O\}$. A finite word $w \in \mathcal{B}^*$ is *realized* at $n \in \mathbb{N}$ when the successive parities of the orbit of n are exactly the letters of w. Write $J^{|w|}(n)$ for the endpoint after those letters, and $\#O(w)$ for the number of odd letters.

The identities in this section are formalized in Lean (`follows_iff_word`, `image_eq_iterate`, `image_append`, `image_monotone_of_follows`, `power_bound_word`, `power_bound_contracts`, `global_defect_identity`, `global_defect_eq_zero_iff_localsTight`, `global_defect_append`, `global_defect_eq_zero_implies_monochrome`, `power_bound_eq_iff_extremal`, `image_eq_start_defectRati`
`one_plus_eta_lt_succ_sq`). The proofs below are the ordinary integer arguments.

Theorem 2.1 (fixed-word monotonicity). If $n \le m$ and both realize w, then $J^{|w|}(n) \le J^{|w|}(m)$.

Proof. Induct on w. The empty word is immediate. For the induction step, the two current states realize the same next letter. On the even branch, monotonicity follows from monotonicity of the integer square root; on the odd branch it follows from monotonicity of $x \mapsto x^3$ followed by the integer square root. Apply the induction hypothesis to the prefix and then the appropriate branch inequality. □

The realizing set of a fixed word need not be an interval.

Theorem 2.2 (finite-word power envelope). If w is realized at n and $m = J^{|w|}(n)$, then

$$m^{2^{|w|}} \le n^{3^{\#O(w)}}.$$

Proof. Write $k = |w|$ and $o = \#O(w)$. The empty word is the equality $n \le n$. Suppose the bound holds at a realized prefix ending at x, and the next letter is realized.

If the next letter is even, then $J(x) = \lfloor \sqrt{x} \rfloor$, so $J(x)^2 \le x$. Raising the inductive bound to the second power gives

$$J(x)^{2^{k+1}} = (J(x)^2)^{2^k} \le x^{2^k} \le n^{3^o}.$$

The odd count is unchanged.

If the next letter is odd, then $J(x) = \lfloor x^{3/2} \rfloor$, so $J(x)^2 \le x^3$. Therefore

$$J(x)^{2^{k+1}} = (J(x)^2)^{2^k} \le x^{3 \cdot 2^k} = (x^{2^k})^3 \le \left(n^{3^o}\right)^3 = n^{3^{o+1}}.$$

This is the claimed bound after one more odd letter. ☐

Corollary 2.3 (exponent-gap contraction). If $n \ge 2$, w is realized at n, and $3^{\#O(w)} < 2^{|w|}$, then $J^{|w|}(n) < n$.

Proof. Let $m = J^{|w|}(n)$ and $k = |w|$. Theorem 2.2 gives $m^{2^k} \le n^{3^{\#O(w)}}$. The exponent gap and $n \ge 2$ give $n^{3^{\#O(w)}} < n^{2^k}$, so $m^{2^k} < n^{2^k}$. Since $m \ge 1$, one has $m < n$. ☐

The corollary includes familiar contracting blocks such as *OOOEE*. It does not prove that every start realizes some contracting word.

The floor slack is exact, and it is not an additive path sum. For a single branch,

$$x^e = J(x)^2 + \rho(x), \qquad e = \begin{cases} 1, & x \text{ even}, \\ 3, & x \text{ odd}, \end{cases}$$

with $0 \le \rho(x) < 2J(x) + 1$. Write

$$\operatorname{gap}(a, \rho, e) = (a + \rho)^e - a^e,$$

so $a^e + \operatorname{gap}(a, \rho, e) = (a + \rho)^e$. If $e \ge 1$, then $\operatorname{gap}(a, \rho, e) = 0$ if and only if $\rho = 0$, and $\rho^e \le \operatorname{gap}(a, \rho, e)$.

Along a realized word w, write $x_0 = n$ and $x_{j+1} = J(x_j)$, and let $\rho_j = \rho(x_j)$. Define a running slack by $D_0 = 0$ and

$$D_{j+1} = \begin{cases} D_j + \operatorname{gap}(x_{j+1}^2, \rho_j, 2^j), & \text{next letter even}, \\ \operatorname{gap}(x_{j+1}^2, \rho_j, 2^j) + \operatorname{gap}(x_j^{2^j}, D_j, 3), & \text{next letter odd}. \end{cases}$$

The *global defect* is $\Delta_w(n) = D_{|w|}$. An even letter keeps the old slack and lifts the new remainder through 2^j. An odd letter first cubes the running slack and then lifts the new remainder. In particular Δ is not the sum of the local remainders.

Theorem 2.4 (global defect identity). If w is realized at n and $m = J^{|w|}(n)$, then

$$n^{3^{\#O(w)}} = m^{2^{|w|}} + \Delta_w(n), \qquad \Delta_w(n) \ge 0.$$

Theorem 2.2 is the inequality $\Delta_w(n) \ge 0$.

Proof. The empty word is $n = n + 0$. Suppose a realized prefix of length k with odd count o ends at x and satisfies $n^{3^o} = x^{2^k} + D$.

If the next letter is even, then $x = J(x)^2 + \rho(x)$, so

$$n^{3^o} = (J(x)^2 + \rho(x))^{2^k} + D = J(x)^{2^{k+1}} + D + \mathrm{gap}(J(x)^2, \rho(x), 2^k).$$

The odd count is unchanged, and the new slack is the even update.

If the next letter is odd, then $x^3 = J(x)^2 + \rho(x)$. Cubing the inductive identity gives

$$n^{3^{o+1}} = (x^{2^k} + D)^3 = x^{3\cdot 2^k} + \mathrm{gap}(x^{2^k}, D, 3).$$

The leading term is

$$x^{3\cdot 2^k} = (J(x)^2 + \rho(x))^{2^k} = J(x)^{2^{k+1}} + \mathrm{gap}(J(x)^2, \rho(x), 2^k),$$

which is the odd update. □

Theorem 2.5 (vanishing). If w is realized at n, the following are equivalent.

1. $\Delta_w(n) = 0$.
2. Every local remainder along w vanishes.
3. $J^{|w|}(n)^{2^{|w|}} = n^{3^{\#O(w)}}$.

In that case w is monochrome. More precisely, either $w = E^k$ and $n = a^{2^k}$ for an even a, or $w = O^k$ and $n = a^{2^k}$ for an odd a. A realized mixed word therefore has $\Delta_w(n) > 0$.

Proof. The identity of Theorem 2.4 gives (1)⇔(3). For (1)⇔(2), use the recurrence. If $e \geq 1$, a power-gap vanishes if and only if its addend vanishes. The even update is a sum of two nonnegative terms, and the odd update is a sum of two power-gaps; either vanishes only when the incoming slack and the new remainder both vanish. Thus $D_{|w|} = 0$ forces $D_0 = 0$ and every $\rho_j = 0$. The converse is immediate.

If every remainder vanishes, an even step satisfies $x_j = x_{j+1}^2$, while an odd step satisfies $x_j^3 = x_{j+1}^2$. In either case x_j and x_{j+1} have the same parity, so the itinerary is monochrome. If $w = E^k$, repeated substitution gives $n = x_k^{2^k}$; writing $a = x_k$, its parity is even. If $w = O^k$, unique factorization applied to $x_j^3 = x_{j+1}^2$ shows that every prime valuation of x_j is divisible by 2. Iterating this valuation relation shows that every prime valuation of $n = x_0$ is divisible by 2^k, hence $n = a^{2^k}$ for an odd a. Conversely, these even and odd towers make every local remainder zero. This is the extremal classification `power_bound_eq_iff_extremal`. □

Theorem 2.6 (composition). If u is realized at n and v is realized at $m = J^{|u|}(n)$, then

$$\Delta_{uv}(n) = \mathrm{gap}(m^{2^{|u|}},\, \Delta_u(n),\, 3^{\#O(v)}) + \mathrm{gap}(J^{|v|}(m)^{2^{|v|}},\, \Delta_v(m),\, 2^{|u|}).$$

In particular $\Delta_v(m)^{2^{|u|}} \leq \Delta_{uv}(n)$. Every local remainder satisfies $\rho_j^{2^j} \leq \Delta_w(n)$.

Proof. Write $o_u = \#O(u)$ and $o_v = \#O(v)$. Theorem 2.4 on u and on uv gives

$$n^{3^{o_u+o_v}} = \left(m^{2^{|u|}} + \Delta_u(n)\right)^{3^{o_v}} = J^{|uv|}(n)^{2^{|uv|}} + \Delta_{uv}(n).$$

The first power expands as

$$m^{2^{|u|}\cdot 3^{o_v}} + \mathrm{gap}(m^{2^{|u|}},\, \Delta_u(n),\, 3^{o_v}).$$

Theorem 2.4 on v at m rewrites the leading term as

$$\left(J^{|v|}(m)^{2^{|v|}} + \Delta_v(m)\right)^{2^{|u|}},$$

which expands as the second gap plus $J^{|uv|}(n)^{2^{|uv|}}$. Comparing the two expressions for $n^{3^{o_u+o_v}}$ yields the identity. The suffix inequality is $\rho^e \le \mathrm{gap}(a,\rho,e)$ on the second summand. For a local remainder at index j, split w after j letters: the suffix defect is at least ρ_j, and raising through 2^j gives the stated bound. □

Composition is polynomial, not additive: later odd letters raise the prefix slack to the third power, and later letters of either parity raise a suffix slack through $2^{|u|}$. The naive recurrence $\Delta \leftarrow \Delta + \rho$ is false.

Corollary 2.7 (cycle surplus). If w is a cycle word at n, that is $J^{|w|}(n) = n$, then

$$\Delta_w(n) = n^{3^{\#O(w)}} - n^{2^{|w|}}$$

exactly (`image_eq_start_defectRatio`). A cycle burns its entire formal surplus in floor losses.

Proof. Theorem 2.4 with $m = n$. Nonnegativity of $\Delta_w(n)$ makes the difference a genuine one. □

The identity does not supply a state-independent positive tax, and none exists: the single-step slack is provably squeezed by the scale of the state. For one realized letter at x with branch exponent $e \in \{1, 3\}$,

$$x^e < (J(x) + 1)^2$$

(`one_plus_eta_lt_succ_sq`), so the relative slack $1 + \eta = x^e/J(x)^2$ satisfies

$$\eta < \frac{2}{J(x)} + \frac{1}{J(x)^2},$$

which tends to 0 as the state grows. A uniform per-step tax is therefore impossible. The recorded extreme is an *OOE* block at $n = 180370579261640036336071806107777 \approx 1.80 \cdot 10^{32}$ whose relative slack satisfies $0 < q < 10^{-30}$; this is an exact-integer computation, not an estimate. The inequality $\Delta_w(n) > n^{3^{\#O(w)}} - n^{2^{|w|}}$ on a formally expanding word is exactly $J^{|w|}(n) < n$, so it does not forbid mixed expanding prefixes.

3. Inverse cells and cycles

The exact one-step fibers are

$$J(n) = q \iff q^2 \le n < (q + 1)^2 \quad (n \text{ even})$$

and

$$J(n) = m \iff m^2 \le n^3 < (m + 1)^2 \quad (n \text{ odd}).$$

An odd fiber contains at most one integer (`odd_cell_unique`). Indeed, suppose $a < b$ lie in the same odd cell indexed by m. Then $(a + 1)^3 \le b^3 < (m + 1)^2$ and $m^2 \le a^3$. Subtracting the latter lower bound from the former upper bound gives

$$3a^2 + 3a + 1 = (a + 1)^3 - a^3 < (m + 1)^2 - m^2 = 2m + 1,$$

and hence $3a^2 \le 2m$. If $a = 0$, then $m = 0$ and the displayed strict inequality is already impossible. If $a > 0$, squaring and using $m^2 \le a^3$ gives $9a^4 \le 4m^2 \le 4a^3$, contradicting $4a^3 < 9a^4$. An even fiber is a parity-restricted square interval and may contain many predecessors.

A nonempty realized word w with $J^{|w|}(n) = n$ is a cycle word.

The identities in this section are formalized in Lean (`cycle_word_formally_expanding`, `cycleMin_not_end_odd`, `square_scale_superquadratic`, `cycleMin_to_even_superquadratic`, `no_cycle_word_oooeoe`, `no_cycle_word_ooooee`, `no_cycle_word_length_le_six` with components `no_cycle_word_replicate_odd`, `cycleWord_rotateWord`, `cycleWord_exists_even_terminating`, `no_cycle_word_length_four_ends_even`, `no_cycle_word_length_five_ends_even`, `no_cycle_word_ooe`, `no_cycle_odd_run_append_even`, `no_cycle_word_ooeooe`). The proofs below are the ordinary integer arguments.

Theorem 3.1 (cycle restrictions). Let w be a cycle word at $n \geq 2$.

(i) The word is formally expanding:
$$2^{|w|} < 3^{\#O(w)}$$
(`cycle_word_formally_expanding`). A contracting word cannot close a nontrivial cycle.

(ii) The cycle minimum is odd and the cycle maximum is even. A minimum-based orientation cannot end in an odd letter (`cycleMin_not_end_odd`).

(iii) A realized word v is *superquadratic* if
$$3^{\#O(v)} \geq 2^{|v|+1}.$$
Any realized path from a start $n \geq 2$ to a state at least n^2 is superquadratic (`square_scale_superquadratic`). On a cycle minimum the path to any later even state is superquadratic (`cycleMin_to_even_superquadratic`). The prefix OOE is expanding ($9 > 8$) but not superquadratic ($9 < 16$), so it cannot carry the minimum to square scale.

Proof. For (i), Theorem 2.2 applied to a cycle endpoint gives $n^{2^{|w|}} \leq n^{3^{\#O(w)}}$. Since $n \geq 2$, comparison of exponents gives $2^{|w|} \leq 3^{\#O(w)}$; equality is impossible because the two sides have different prime divisors and $|w| \geq 1$.

Every state on such a cycle is at least 2: once an orbit reaches 1, it remains there and cannot return to a start $n \geq 2$. An even state $x \geq 2$ satisfies $J(x) < x$, so a cycle minimum cannot be even. An odd state $x \geq 3$ satisfies $J(x) > x$, so a cycle maximum cannot be odd. This proves the first assertion of (ii). For the final-letter assertion, let x be the predecessor of a minimum-oriented return n. If the last letter were odd, the odd return cell would give $n^2 \leq x^3 < (n+1)^2$. Minimality gives $x \geq n$, hence $n^3 < (n+1)^2$, impossible for $n \geq 3$; and the minimum is odd, so $n \geq 3$.

For (iii), let a realized word v send n to $y \geq n^2$. Theorem 2.2 gives
$$n^{2^{|v|+1}} = (n^2)^{2^{|v|}} \leq y^{2^{|v|}} \leq n^{3^{\#O(v)}}.$$

Since $n \geq 2$, comparison of exponents proves the superquadratic inequality. On a cycle minimum, if a later state y is even, its successor is at least n. Thus $n \leq J(y) = \lfloor \sqrt{y} \rfloor$, so $n^2 \leq y$, and the preceding argument applies to that prefix. □

These restrictions assemble into a complete census of short cycles. Rotation reduces any cycle word to an even-terminating orientation, the expanding filter of Theorem 3.1(i) and threshold arguments eliminate every even-terminating word of length at most six except $OOOEOE$ and $OOOOEE$, and those two survivors require an individual argument, which uses the last-even cell together with a coarse lower envelope. That argument is Lemma 3.2; the census is Theorem 3.3. No exclusion of cycles of length seven or more is claimed.

For $n \geq 1$,

$$n < 4\,\lfloor\sqrt{n}\rfloor^2,$$

so an even step satisfies $n \leq 4\,J(n)^2$ and an odd step satisfies $n^3 \leq 4\,J(n)^2$. Composing along a realized word v of length k with odd count o gives

$$n^{3^o} \leq D_v\, J^k(n)^{2^k}.$$

The denominator starts at 1 and updates by $D \mapsto D \cdot 4^{2^j}$ on an even letter and $D \mapsto D^3 \cdot 4^{2^j}$ on an odd letter, at step j. In particular $D_{OOO} = 2^{38}$ and $D_{OOOO} = 2^{130}$. If a cycle word ends in an even letter, the last-even cell is $J^{|w|-1}(n) < (n+1)^2$.

Lemma 3.2 (two length-six exclusions). Neither *OOOEOE* nor *OOOOEE* is a cycle word at any $n \geq 2$.

Proof. First, if $n \geq 256$, then

$$n^{81} > 2^{130}(n+1)^{64}.$$

Indeed $257^{64} < 2 \cdot 256^{64}$ and $256(n+1) \leq 257\,n$, so $(n+1)^{64} < 2n^{64}$. Then $2^{130}(n+1)^{64} < 2^{131}n^{64}$. Since $n \geq 256 = 2^8$, one has $n^{17} \geq 2^{136}$, and therefore $2^{131}n^{64} < 2^{136}n^{64} \leq n^{81}$.

For $n < 256$, neither word returns to its start. Lean checks both itinerary-and-return tables by `native_decide` on `Fin 256`; the same mechanism also verifies the finite numerical inequality $257^{64} < 2 \cdot 256^{64}$ used at the cutoff.

Now suppose $n \geq 256$ realizes *OOOOEE*, and write $z = J^4(n)$ for the image after the prefix *OOOO*. The last two even letters give $J(z) < (n+1)^2$ and $z < (J(z)+1)^2$, hence $z < (n+1)^4$. The lower envelope on *OOOO* is $n^{81} \leq 2^{130}z^{16}$, so $n^{81} < 2^{130}(n+1)^{64}$, contradicting the tail inequality.

Finally suppose $n \geq 256$ realizes *OOOEOE*. Write $z_3 = J^3(n)$ and $y = J(z_3) = \lfloor\sqrt{z_3}\rfloor$, so $z_3 < (y+1)^2$. The lower envelope on *OOO* is $n^{27} \leq 2^{38}z_3^8 < 2^{38}(y+1)^{16}$. Cubing yields $n^{81} < 2^{114}(y+1)^{48}$. The last letters *OE* give the odd-cell bound $y^3 < (n+1)^4$. Write $A = n+1 \geq 257$. Then $(y+1)^3 < 2A^4$: if $y \leq A$, this follows from $(A+1)^3 < 2A^4$; if $y > A$, then $y \geq 4$ and $(y+1)^3 = y^3 + 3y^2 + 3y + 1 < A^4 + 4y^2$, while $4y^2 < A^4$ because $4y^3 < 4A^4 \leq yA^4$. Raising $(y+1)^3 < 2A^4$ to the sixteenth power gives $(y+1)^{48} < 2^{16}(n+1)^{64}$. Combining with the cubed lower envelope produces again $n^{81} < 2^{130}(n+1)^{64}$. $\square$

Theorem 3.3 (small-cycle census). No word of length at most six is a cycle word at any $n \geq 2$ (`no_cycle_word_length_le_six`). Equivalently, a nontrivial Juggler cycle, if one exists, has period at least seven.

Proof. Rotating a cycle word by one letter moves the base point one step along the orbit and yields another cycle word (`cycleWord_rotateWord`); every state of the cycle is at least 2, because an orbit that reaches 1 stays at 1 and cannot return to a start $n \geq 2$.

If every letter is odd, the start is odd, hence $n \geq 3$, and the odd branch strictly increases there: $J(x) > x$ for odd $x \geq 3$, since $x^3 \geq (x+1)^2$. The orbit ascends strictly and never returns (`no_cycle_word_replicate_odd`). Otherwise some letter is even, and a rotation ending just after that letter produces an even-terminating cycle word vE of the same length based at a cycle state $m \geq 2$ (`cycleWord_exists_even_terminating`). It therefore suffices to exclude even-terminating cycle words of length at most six.

By Theorem 3.1(i) a cycle word is formally expanding. No even-terminating word of length one or two is expanding ($2 < 3^0$ and $4 < 3$ both fail). For length three the only expanding candidate is OOE: for $m \geq 5$ the next-square threshold after OO gives $J^2(m) \geq (m+1)^2$, contradicting the last-even cell $J^2(m) < (m+1)^2$; the only smaller odd start is $m = 3$, where $J^2(3) = 11$ is odd while the final even letter requires an even state. For length four the expanding filter leaves only O^3E, and for length five only O^4E; the threshold inherits along appended odd letters, and the same last-even contradiction applies (`no_cycle_word_length_four_ends_even`, `no_cycle_word_length_five_ends_even`).

For length six the filter requires at least four odd letters among the first five, leaving O^5E, $EOOOOE$, $OEOOOE$, $OOEOOE$, $OOOEOE$, and $OOOOEE$. The odd-run threshold excludes O^5E (`no_cycle_odd_run_append_even`). For $OOEOOE$, a minimum-based orientation meets the same next-square threshold at the internal even letter after the prefix OO, with starts below 5 checked directly (`no_cycle_word_ooeooe`). $EOOOOE$ rotates one step onto $OOOOEE$, and $OEOOOE$ rotates two steps onto $OOOEOE$. The two remaining words are excluded by Lemma 3.2. □

The census stops at length six because length seven admits even-terminating expanding words with two internal even letters that none of the recorded thresholds reach; no exclusion at length seven is claimed.

4. Short descent certificates

A start $n \geq 2$ has a *descent certificate* if there exists a realized finite word w with $J^{|w|}(n) < n$, or with image 1. Lean packages this predicate as `FiniteProgress`, an abbreviation of `DescentCertificate`. The four constructors of that type are proof forms of the same predicate, not four different claims. The predicate is existential over all finite words, not only over words of length one or two.

The theorems in this section are formalized in Lean (`even_finiteProgress`, `odd_even_finiteProgress`, `unresolved_is_odd_odd`, `reachesOne_of_all_finiteProgress`, `reachesOne_of_lt_twelve`, `even_lt_sq_twelve_reachesOne`). The proofs below are the ordinary integer arguments. Proposition 4.4 is the exact census of Section 1.1, not a Lean theorem.

Theorem 4.1 (uniform short certificates). Let $n \geq 2$.

1. If n is even, then the one-letter word E is a descent certificate: $J(n) = \lfloor \sqrt{n} \rfloor < n$.
2. If n is odd and $J(n)$ is even, then the two-letter word OE is a descent certificate.

Proof. For (1), $\lfloor \sqrt{n} \rfloor \leq \sqrt{n} < n$ for $n \geq 2$. For (2), the word OE is realized by hypothesis, and

$$J^2(n) = \lfloor \sqrt{\lfloor n^{3/2} \rfloor} \rfloor \leq n^{3/4} < n$$

for $n \geq 2$. □

Theorem 4.2 (unresolved starts are odd-to-odd). If $n \geq 2$ has no descent certificate, then n is odd and $J(n)$ is odd.

Proof. Contrapositive of Theorem 4.1: an even start and an odd-to-even start each carry a short certificate. □

The converse is false: many odd-to-odd starts descend after a longer word.

If every start above 1 has some descent certificate, ordinary strong induction yields arrival at 1 (`reachesOne_of_all_finiteProgress`). The hypothesis is not proved. A certificate at n only reduces the problem to a strictly smaller positive integer, which may itself be odd-to-odd.

Lean also certifies a finite landing class:

Theorem 4.3 (small residuals). Every $y \in \{1, \dots, 11\}$ reaches 1. Consequently every even residual strictly below $144 = 12^2$ reaches 1.

Proof. The orbits are finite and merge quickly: $2 \to 1$, $4 \to 2$, $6 \to 2$, $8 \to 2$, $3 \to 5 \to 11 \to 36 \to 6$, $7 \to 18 \to 4$, $9 \to 27 \to 140 \to 11$, and $10 \to 3$. Each chain ends on a start already settled, so every $y \le 11$ reaches 1. For the second claim, an even $y < 144$ has $J(y) = \lfloor \sqrt{y} \rfloor \le 11$, which lands in the verified set. (Lean evaluates the same finite orbits in `reachesOne_of_lt_twelve` and `even_lt_sq_twelve_reachesOne`.) $\square$

This enlarges the set of fatal landings for a hypothetical minimal counterexample.

On the complementary odd-to-odd class, first return below the start is frequent at a short horizon, but not automatic.

Proposition 4.4 (odd-to-odd first-return census). For starts $2 \le n \le N$ and horizon 20, write OO for the odd-to-odd starts in that range. The exact census is:

N	#OO	OO first return ≤ 20	all starts, first return ≤ 20
10^3	252	221/252 (0.877)	968/999 (0.969)
10^4	2504	2220/2504 (0.887)	9715/9999 (0.972)
10^5	24984	22379/24984 (0.896)	97394/99999 (0.974)
10^6	249926	223683/249926 (0.895)	973756/999999 (0.974)

At $N = 10^6$, 26,243 odd-to-odd starts have no return below the start in 20 steps. These rows use exact Python integer arithmetic with no size cutoff through step 20; the unresolved count is zero in every row. They are not Lean-certified and do not constitute an almost-all theorem.

5. Parity discrepancy and certified-descent densities

This section counts itinerary word classes. It proves the ambient depth-1 estimate (Theorem 5.1) with its density-3/4 corollary, then extends the method through itinerary depth four by exact linearization of the nested floor powers: depth 2 and 3 completely (Theorem 5.4, Proposition 5.5), and depth 4 completely except the *OOO*$*$ split (Theorems 5.7 and 5.8), giving the certified-descent density 13/16 (Corollary 5.9). All estimates here are human proofs; the exact floor reductions beneath them are Lean-verified (Section 1.1).

For odd n write $s(n) = (-1)^{\lfloor n^{3/2} \rfloor}$ and

$$S_O(N) = \sum_{\substack{n \le N \\ n \text{ odd}}} s(n).$$

Let $M(N)$ be the number of odd integers $n \le N$, and let

$$\mathrm{OO}(N) = \#\{n \le N : \ n \text{ odd}, \ J(n) \text{ odd}\}.$$

Then $s(n) = -1$ if and only if $J(n)$ is odd, and

$$S_O(N) = M(N) - 2\,\mathrm{OO}(N).$$

Also $M(N) = N/2 + O(1)$.

The floor sign has an exact fractional-part form. Write $x = \lfloor x \rfloor + \{x\}$. If $\lfloor x \rfloor$ is even then $\{x/2\} < 1/2$; if $\lfloor x \rfloor$ is odd then $\{x/2\} \geq 1/2$. Thus

$$\lfloor x \rfloor \text{ is odd} \quad \Longleftrightarrow \quad \{x/2\} \geq \tfrac{1}{2}$$

(Lean: `floor_odd_iff_half_le_fract_half`). For odd $n = 2r + 1$ set

$$g(r) = \frac{(2r+1)^{3/2}}{2}.$$

Then $s(n) = -1$ if and only if $\{g(r)\} \geq 1/2$. For R points $x_0, \dots, x_{R-1}$ and an interval $I \subset [0, 1)$, define the unnormalized interval discrepancy

$$\mathcal{D}_R(I) = \#\{0 \leq r < R : \{x_r\} \in I\} - R|I|.$$

Taking $R = M(N)$, the odd integers at most N are exactly $2r + 1$ for $0 \leq r < R$, and therefore $S_O(N) = -2\mathcal{D}_R([1/2, 1))$ for $x_r = g(r)$.

The following two lemmas are classical. The derivative estimate is the $k = 2$ case of Iwaniec–Kowalski [11, Thm. 8.20, Sec. 8.3, pp. 204–213]. The discrepancy inequality is Kuipers–Niederreiter [10, Ch. 2, Sec. 2, Thm. 2.5, pp. 112–115].

Lemma 5.A (van der Corput, second-derivative form). Let f be twice differentiable on an interval of length M, with $\lambda \leq |f''| \leq \alpha\lambda$ for some $\alpha \geq 1$. Then

$$\Big|\sum e(f)\Big| \ll_\alpha M\lambda^{1/2} + \lambda^{-1/2},$$

where the sum runs over the integers in the interval and $e(t) = e^{2\pi i t}$.

Lemma 5.B (Erdős–Turán). The discrepancy of R points $x_j \in \mathbb{R}/\mathbb{Z}$ against an interval satisfies

$$D \ll \frac{R}{H} + \sum_{h=1}^{H} \frac{1}{h} \Big|\sum_{j=1}^{R} e(hx_j)\Big|$$

for every cutoff $H \geq 1$.

Theorem 5.1 (ambient odd-input discrepancy).

$$|S_O(N)| \ll N^{5/6}.$$

This argument is not packaged in Lean.

Proof. The second derivative is

$$g''(r) = \tfrac{3}{2}(2r+1)^{-1/2}.$$

It is positive and decreasing. Separate the initial term $r = 0$, then partition $1 \leq r < R$ into blocks $[M, \min(2M, R))$, where M runs over powers of 2. On such a block $g''(r) \asymp M^{-1/2}$. For the h-th Fourier mode take $f = hg$, so $\lambda \asymp hM^{-1/2}$. Lemma 5.A gives

$$\Big|\sum_{r \asymp M} e(hg(r))\Big| \ll h^{1/2}M^{3/4} + h^{-1/2}M^{1/4}.$$

Lemma 5.B, with $R \asymp M$, bounds that block's contribution to $S_O(N)$ by

$$\frac{M}{H} + M^{3/4}H^{1/2} + M^{1/4}.$$

Here the middle term is $\sum_{h \le H} h^{-1} \cdot h^{1/2}M^{3/4} \ll M^{3/4}H^{1/2}$, and the last term is $\sum_{h \le H} h^{-1} \cdot h^{-1/2}M^{1/4} \ll M^{1/4}$. For bounded M, the trivial estimate is absorbed into the constant. Otherwise choose $H = \lfloor M^{1/6} \rfloor$, which balances the first two terms at $M^{5/6}$. The initial term contributes $O(1)$, and the geometric sum of the block bounds is $\sum_{M \le R} O(M^{5/6}) = O(R^{5/6}) = O(N^{5/6})$. The floor sign is not replaced by a single exponential; the exponential sums are those of the sequence $\{g(r)\}$. $\square$

Corollary 5.2 (short-certificate class). The odd-to-odd starts have natural density 1/4:

$$\left|\mathrm{OO}(N) - N/4\right| \ll N^{5/6}.$$

Equivalently, the starts that admit the uniform one- or two-step certificates of Theorem 4.1 — every even start together with every odd-to-even start — have natural density 3/4.

This is a density of the uniform short-certificate class of Theorem 4.1, not of all descent certificates and not of starts that reach 1.

The exact census through $N = 10^7$ has $\mathrm{OO}(N)/N = 0.2499896$. Through $N = 10^6$ one has $S_O(N) = 146$ and running maximum 256, at $n = 985351$. A spot computation at 10^7 has running maximum 459. The observed growth is much smaller than $N^{5/6}$: the recorded maxima at 10^6 and 10^7 sit near the scale $N^{1/3}$. That empirical scale is an observation, not a theorem.

Theorem 5.1 concerns consecutive source intervals. It supplies no transfer theorem for orbit samples or sparse image sets, so it cannot by itself control the odd-to-odd dynamics after the first step. The remainder of this section does not transfer Theorem 5.1 to image sets; it attacks the nested sums directly, by removing the inner floors exactly before any analytic step.

For the rest of the section, n is odd, $X = n^{3/2}$, $m = \lfloor X \rfloor = \mathrm{isqrt}(n^3)$, and $\theta = X - m \in [0, 1)$. At the second level, $Y = m^{3/2}$, $v = \lfloor Y \rfloor$, $\theta_2 = Y - v$; on the *OE* branch (m even), $U = m^{1/2}$, $w = \lfloor U \rfloor = J^2(n)$, and $\theta_w = U - w$. Write $\psi(x) = (-1)^{\lfloor x \rfloor}$. The first image parity of an odd start is $(-1)^m = \psi(n^{3/2})$; the next letters are the parities of v (after an odd image) or w (after an even image). Each ψ-factor below is evaluated unconditionally; the indicator algebra of the corollaries only ever weights it on the branch where it computes the true itinerary letter, and these branch-consistency identities are machine-checked exactly on verification windows.

Lemma 5.3 (exact linearization and gap cells). (i) For every odd $n \ge 3$,

$$m^{3/2} = \tfrac{3}{2}\, m\, n^{3/4} - \tfrac{1}{2}\, n^{9/4} + E(n), \qquad 0 \le E(n) \le \tfrac{3}{8}\,(n^{3/2} - 1)^{-1/2} \le \tfrac{1}{2}\, n^{-3/4}.$$

(ii) For $h \ge 1$ and odd n, let $g(n) = m(n + 2h) - m(n)$ and $\delta(n) = (n + 2h)^{3/2} - n^{3/2}$. Then

$$g(n) = \lfloor \delta(n) \rfloor + \kappa(n), \qquad \kappa(n) = \left[\{n^{3/2}\} \ge 1 - \{\delta(n)\}\right] \in \{0, 1\}.$$

Proof. (i) Let $f(t) = (X - t)^{3/2}$ on $[0, \theta]$, so $f(\theta) = m^{3/2}$. Taylor with Lagrange remainder at 0 gives $f(\theta) = X^{3/2} - \frac{3}{2}X^{1/2}\theta + \frac{3}{8}(X - \xi)^{-1/2}\theta^2$ for some $\xi \in (0, \theta)$. Substituting $\theta = X - m$ in the linear term yields $-\frac{1}{2}X^{3/2} + \frac{3}{2}mX^{1/2} = -\frac{1}{2}n^{9/4} + \frac{3}{2}mn^{3/4}$, and the remainder lies in $[0, \frac{3}{8}(X - 1)^{-1/2}]$, with $(X - 1)^{-1/2} \le \frac{4}{3}X^{-1/2}$ already for $n \ge 3$. (ii) $g = \lfloor X + \delta \rfloor - \lfloor X \rfloor = \lfloor \delta \rfloor + \lfloor \{X\} + \{\delta\} \rfloor$, and the last floor is 1 precisely when $\{X\} + \{\delta\} \ge 1$ (Lean: `floor_gap_eq_carry`, `seq_floor_gap`). $\square$

The point of (i): the non-smooth integer m enters the phase *linearly*, multiplied by a smooth coefficient; no fractional part of growing amplitude survives. The naive expansion $m^{3/2} = n^{9/4} - \frac{3}{2}\theta n^{3/4} + O(n^{-3/4})$ instead leaves the sawtooth θ with the growing amplitude $n^{3/4}$, which defeats the van der Corput method directly. For (ii), since δ is smooth and increasing with $\delta' \asymp hn^{-1/2}$, the level sets of $\lfloor\delta\rfloor$ partition a dyadic block $n \in (P, 2P]$ into $O(hP^{1/2})$ cells of length $\asymp P^{1/2}/h$ on which the integer part of the gap is constant.

Theorem 5.4 (nested parity discrepancy). For every $\varepsilon > 0$ and every $(a, b) \in \{0,1\}^2 \setminus \{(0,0)\}$,

$$\Big|\sum_{n\le N,\ n \text{ odd}} \psi(n^{3/2})^a\, \psi(m(n)^{3/2})^b\Big| \ll_\varepsilon N^{23/24+\varepsilon}.$$

Consequently each of the four joint parity classes of $(m, \lfloor m^{3/2}\rfloor)$ on odd $n \le N$ has cardinality $\frac{N}{8} + O(N^{23/24+\varepsilon})$; in particular the odd-to-odd cylinder refines one letter deeper than Theorem 5.1.

Proof. Work on dyadic blocks $n \in (P, 2P]$, odd, with truncations $J_1 = J_2 = P^{1/24}$ (wave modes), $H = P^{1/12}$ (differencing), and $R = P^{1/4}$ (cell modes).

Step 1 (wave expansion). By Vaaler's theorem [13] applied to the period-2 square wave, $\psi(x) = V_J(x/2) + O(\Delta_J(x/2))$, where $V_J(t) = \sum_{0<|q|\le J} a_q e(qt)$ with $|a_q| \le \min(1, 2/|q|)$ and $\Delta_J \ge 0$ is a trigonometric polynomial of degree J with constant term and coefficients $O(1/J)$. For the product, $|\psi_1\psi_2 - V_1V_2| \le \Delta_1 + \Delta_2 + \Delta_1\Delta_2$, and each error term is again a nonnegative trigonometric polynomial of the same shape. Expanding both factors reduces the theorem to bounds, uniform in the mode pair, for

$$S_{\mu,\nu}(P) = \sum_{n\in(P,2P],\ n \text{ odd}} e(\mu\, n^{3/2} + \nu\, m^{3/2}), \qquad |\mu| \le J_1,\ \tfrac{1}{2} \le |\nu| \le J_2,$$

with $\mu, \nu \in \frac{1}{2}\mathbb{Z}$; the pure first-factor majorant sums are $\ll P/J_1 + P^{5/6+\varepsilon} \ll P^{23/24+\varepsilon}$ by the depth-1 machinery, and mode weights sum to $O(\log^2 P)$. Write $i = 2\mu$, $j = 2\nu$.

Step 2 (linearization). By Lemma 5.3(i), replacing $\frac{j}{2}m^{3/2}$ by $\frac{3j}{4}mn^{3/4} - \frac{j}{4}n^{9/4}$ changes $S_{\mu,\nu}$ by at most $2\pi\frac{j}{4}\sum_{n\sim P} n^{-3/4} \ll jP^{1/4} \le P^{7/24}$.

Step 3 (van der Corput A-process). With $H = P^{1/12}$, $|S_{\mu,\nu}|^2 \le \frac{2P}{H}\sum_{0\le h<H} |T_h|$, where T_h sums $e(\Phi(n{+}2h) - \Phi(n))$ over the overlap and Φ is the linearized phase. The exact rewrite $m(n{+}2h)(n{+}2h)^{3/4} - m(n)n^{3/4} = g(n)(n{+}2h)^{3/4} + m(n)[(n{+}2h)^{3/4} - n^{3/4}]$ with $m(n) = n^{3/2} - \theta_n$ gives

$$\Phi(n{+}2h) - \Phi(n) = A_h(n) + \tfrac{3j}{4}\, g(n)\, (n{+}2h)^{3/4} + O(jhP^{-1/4}),$$

where A_h is smooth and the error is the $-\frac{3j}{4}\theta_n[(n{+}2h)^{3/4} - n^{3/4}]$ term. The two $n^{5/4}$-scale contributions of A_h cancel at leading order, leaving $A_h'' = \frac{81}{256}jh^2n^{-7/4}(1 + o(1))$ plus an i-part $O(ihP^{-3/2})$, both $o(jhP^{-3/4})$ in these ranges.

Step 4 (cells). Split T_h over the $O(hP^{1/2})$ cells of Lemma 5.3(ii). On a cell, $g = G + \kappa$ with G constant and $\kappa(n) = [\{n^{3/2}\} \ge \rho(n)]$, $\rho = 1 - \{\delta(n)\}$ smooth and monotone there. Vaaler-expand κ with modes $e(rn^{3/2})$, $0 < |r| \le R$. The moving endpoint costs nothing: since $\rho(n) = 1 + G - \delta(n)$, the coefficient's n-dependent piece contributes the exact smooth phase $r\delta(n)$, and $e(rn^{3/2} + r\delta(n)) = e(r(n{+}2h)^{3/2})$ up to a unimodular constant. Every r-term therefore falls into one of the two smooth families $e(rn^{3/2})$, $e(r(n{+}2h)^{3/2})$; the main term $\delta(n) - G$ has total variation exactly 1 per cell and is absorbed by one Abel summation.

Step 5 (second-derivative test per cell). The remaining phases are smooth. For the $r = 0$ parts, $|f''| \asymp \lambda_0 = jhP^{-3/4}$ with a single sign and bounded ratio across a cell; van der Corput II [14] on a cell of length $\ell \asymp P^{1/2}/h$ gives $\ll \ell\lambda_0^{1/2} + \lambda_0^{-1/2}$, hence over all cells $\ll (jh)^{1/2}P^{5/8} + j^{-1/2}h^{1/2}P^{7/8}$. For $r \neq 0$, the curvature is dominated by the r-term $\asymp |r|P^{-1/2}$ of fixed sign (as $jh \leq P^{1/8} \ll P^{1/4}$); the same test and the $1/|r|$ weights give $\ll R^{1/2}P^{3/4} + hP^{3/4}\log P$, and the majorant errors contribute $O(P/R) = O(P^{3/4})$. In total $|T_h| \ll P^{7/8}(1 + h^{1/2})$, uniformly in the modes.

Step 6 (assembly). $|S_{\mu,\nu}|^2 \ll P^2/H + P^{15/8}H^{1/2} \ll P^{23/12}$ at $H = P^{1/12}$, so $|S_{\mu,\nu}| \ll P^{23/24}$. Summing mode weights, majorants, and dyadic blocks gives the theorem with $N^{23/24}\log^3 N$. $\square$

The exponent 23/24 is deliberately unoptimized; every stage has slack. The observed cancellation is far stronger (envelope exponent ≈ 0.28 in exact censuses), which the proof does not claim.

Proposition 5.5 (OE-branch third letter). For $a \in \{0, 1\}$,

$$\Big|\sum_{n \leq N,\ n \text{ odd}} \left((-1)^m\right)^a \psi(m^{1/2})\Big| \ll_\varepsilon N^{7/8+\varepsilon},$$

and consequently $\#\mathrm{OEO}(N),\ \#\mathrm{OEE}(N) = N/8 + O(N^{7/8+\varepsilon})$. Together with Theorem 5.4 this makes depth 3 complete: each of *OOO*, *OOE*, *OEO*, *OEE* has density 1/8 with a power saving.

Proof. Taylor expansion of $(X - \theta)^{1/2}$ with both correction terms one-signed gives

$$m^{1/2} = n^{3/4} + D_1(n), \qquad -\tfrac{1}{2}X^{-1/2} - \tfrac{1}{8}(X-1)^{-3/2} \leq D_1 \leq 0.$$

D_1 is decaying, so replacing $\frac{l}{2}m^{1/2}$ by $\frac{l}{2}n^{3/4}$ inside a Vaaler mode costs $\ll l\sum_{n\sim P} n^{-3/4} \ll lP^{1/4}$, absorbable at $l \leq 2P^{1/24}$. The mode sums are then pure: $\sum_{n\sim P} e(\frac{i}{2}n^{3/2} + \frac{l}{2}n^{3/4})$ with $l \neq 0$. For $i \neq 0$, $\varphi'' \asymp |i|P^{-1/2}$ is single-signed and van der Corput II gives $\ll i^{1/2}P^{3/4} + i^{-1/2}P^{1/4}$; for $i = 0$, $\varphi'' \asymp lP^{-5/4}$ gives $\ll l^{1/2}P^{3/8} + l^{-1/2}P^{5/8}$. Vaaler majorants, truncation tails, and dyadic assembly are as in Theorem 5.4, and every bound is $\ll P^{7/8}$ after summing mode weights. Branch consistency: the $(1 + (-1)^m)/2$ factor vanishes exactly on odd m, where J^2 takes the 3/2-power branch, so the unconditional $\psi(m^{1/2})$ is only weighted where it computes the true third letter. $\square$

Lemma 5.6 (fourth-letter linearization). For every odd $n \geq 3$,

$$v^{1/2} = n^{9/8} + D(n), \qquad -\tfrac{3}{4}\,n^{-3/8} - n^{-9/8} \leq D(n) \leq 0.$$

Proof. Two applications of the Lemma 5.3(i) pattern. With $f(t) = (Y - t)^{1/2}$ on $[0, \theta_2]$: $v^{1/2} = m^{3/4} - \frac{1}{2}\theta_2 m^{-3/4} - E_2$, $0 \leq E_2 \leq \frac{1}{8}(Y-1)^{-3/2}$. With $f(t) = (X - t)^{3/4}$ on $[0, \theta]$: $m^{3/4} = n^{9/8} - \frac{3}{4}\theta n^{-3/8} - E_3$, $0 \leq E_3 \leq \frac{3}{32}(X-1)^{-5/4}$. Every term is nonpositive after the leading one, and $\frac{1}{2}m^{-3/4} + E_3 + E_2 \leq n^{-9/8}$ for $n \geq 3$. $\square$

Unlike the depth-2 layer, *every* non-smooth term now has decaying amplitude: the cumulative phase cost of replacing $\frac{k}{2}v^{1/2}$ by the smooth $\frac{k}{2}n^{9/8}$ over a dyadic block is $\ll kP^{5/8}$, negligible against the target.

Theorem 5.7 (triple parity discrepancy). For every $\varepsilon > 0$ and every $(a, b, c) \in \{0,1\}^3 \setminus \{(0,0,0)\}$,

$$\Big|\sum_{n \leq N,\ n \text{ odd}} \psi(n^{3/2})^a\, \psi(m^{3/2})^b\, \psi(v^{1/2})^c\Big| \ll_\varepsilon N^{23/24+\varepsilon}.$$

Consequently each of the eight sign classes of $(\psi(n^{3/2}), \psi(m^{3/2}), \psi(v^{1/2}))$ on odd $n \leq N$ has cardinality $\frac{N}{16} + O(N^{23/24+\varepsilon})$.

Proof. The $c = 0$ cases are Theorem 5.4 and Theorem 5.1. For $c = 1$, wave-expand all present sign factors as in Step 1 of Theorem 5.4, with truncations $J_1 = J_2 = J_3 = P^{1/24}$, and bound

$$S_{i,j,k}(P) = \sum_{n\in(P,2P],\ n \text{ odd}} e\left(\tfrac{i}{2}n^{3/2} + \tfrac{j}{2}m^{3/2} + \tfrac{k}{2}v^{1/2}\right)$$

uniformly for $|i| \le 2J_1$, $|j| \le 2J_2$, $1 \le |k| \le 2J_3$. By Lemma 5.6, replacing $\frac{k}{2}v^{1/2}$ by $\frac{k}{2}n^{9/8}$ costs $\ll |k|P^{5/8} \le P^{2/3}$, before any differencing.

If $j = 0$, the remaining sum is a single smooth exponential sum with phase $\frac{i}{2}n^{3/2} + \frac{k}{2}n^{9/8}$. Both curvature terms are positive for positive modes, and for mixed signs with $|i| \ge 1$ the first dominates by $P^{3/8-1/24}$; van der Corput II gives $\ll P^{5/8}$ for $i = 0$ and $\ll |i|^{1/2}P^{3/4} + P^{1/4}$ otherwise, both $\ll P^{23/24}$.

If $j \ne 0$, the phase is exactly the Theorem 5.4 phase plus the smooth passenger $\frac{k}{2}n^{9/8}$. Steps 2–6 run verbatim; the passenger modifies the smooth part of the differenced phase by $\frac{k}{2}[(n+2h)^{9/8} - n^{9/8}]$, whose second derivative $\ll |k|hP^{-15/8}$ is smaller than the retained cell curvature $jhP^{-3/4}$ and the r-mode curvature $|r|P^{-1/2}$ by powers of P. Every sign-dominance check of Theorem 5.4 holds with these margins. □

Theorem 5.8 (the OE splits).**

$$\#\mathrm{OEOE}(N),\ \#\mathrm{OEOO}(N) = \tfrac{N}{16} + O(N^{7/8+\varepsilon}), \qquad \#\mathrm{OEEE}(N),\ \#\mathrm{OEEO}(N) = \tfrac{N}{16} + O(N^{13/16+\varepsilon}).$$

Together with Theorems 5.4 and 5.7 this proves every depth-4 itinerary word class except $OOO*$.

Proof. On the OE branch m is even and $w = \lfloor U \rfloor$, $U = m^{1/2}$. The w-level linearization is Lemma 5.3(i) verbatim with base U: since $Um^{1/4} = m^{3/4}$ exactly,

$$w^{3/2} = m^{3/4} - \tfrac{3}{2}\, m^{1/4}\, \theta_w + E, \qquad 0 \le E \le \tfrac{3}{8}(U-1)^{-1/2},$$

and $m^{3/4} = n^{9/8}$ up to one-signed corrections of scale $n^{-3/8}$. For a Vaaler mode $k \ne 0$, the phase is therefore $\varphi_1(n) - B(n)\theta_w(n)$ up to absorbable errors, with $\varphi_1 = \frac{i}{2}n^{3/2} + \frac{j}{2}n^{3/4} + \frac{k}{2}n^{9/8}$ and $B = \frac{3k}{4}n^{3/8}(1 + O(n^{-3/2}))$: a *single growing sawtooth* of amplitude $\asymp kn^{3/8}$ riding $\theta_w = \{m^{1/2}\}$, whose underlying integer increments once every $\asymp \frac{4}{3}n^{1/4}$ steps. Truncations: $J_1 = J_2 = J_3 = P^{1/8}$.

Drift-1 intervals. $B' \asymp kn^{-5/8}$, so B drifts by at most 1 on intervals I of length $L_0 = P^{5/8}/k$; there are $\asymp kP^{3/8}$ of them. On I, expand $e(-B\{U\}) = \sum_r a_r(B)e(rU)$ (Fourier in U, Vaaler truncation $|r+B| \le T = P^{5/16}$); the coefficients satisfy $|a_r(B)| \le \min(1, |r+B|^{-1})$ with window mass $O(\log T)$, and their n-dependence through $B(n)$ has total variation $O(1)$ per interval, removed by one partial summation per mode.

Mode sums. Smoothing $rU \to rn^{3/4}$ costs $kP^{5/8}$ in total. Writing $r = -B_0 + t$, $|t| \le T$, the curvature of the mode phase is

$$\varphi'' = \tfrac{27k}{128}n^{-7/8} - \tfrac{3}{16}(t + \tfrac{j}{2})n^{-5/4} + \tfrac{3i}{8}n^{-1/2}.$$

For $i \ne 0$ the last term dominates and van der Corput II gives $\ll i^{1/2}P^{3/4} + i^{-1/2}kP^{5/8}$ after summing intervals. For $i = 0$, $(T + J_2)P^{-5/4} \ll kP^{-7/8}$ because $T = P^{5/16} \ll kP^{3/8}$ for every $k \ge 1$, so $\lambda \asymp kn^{-7/8}$ is single-signed; van der Corput II per interval and summation over $\asymp kP^{3/8}$ intervals give $S_k \ll k^{1/2}P^{13/16+\varepsilon}$. With Vaaler weights $1/k$, $\sum_{k\le J_3} k^{-1/2}P^{13/16} = J_3^{1/2}P^{13/16} = P^{7/8}$, balanced against the truncation error $P/J_3 = P^{7/8}$ at $J_3 = P^{1/8}$. Total $\ll P^{7/8+\varepsilon}$.

OEE branch. The fourth letter needs $\psi(w^{1/2})$, and $w^{1/2} = U^{1/2} - \frac{1}{2}\theta_w U^{-1/2} - \frac{1}{8}\theta_w^2(U-\xi)^{-3/2}$ has *decaying* amplitudes only, with $U^{1/2} = m^{1/4} \to n^{3/8}$ likewise. The pure modes $e(\frac{l}{2}n^{3/8})$ have $\lambda \asymp lP^{-13/8}$, giving $l^{1/2}P^{3/16} + l^{-1/2}P^{13/16}$; the mixed cases sit in the dominance hierarchy $iP^{-1/2} \gg jP^{-5/4} \gg lP^{-13/8}$ with all three second derivatives of the same sign. Total $\ll N^{13/16+\varepsilon}$. $\square$

Corollary 5.9 (depth-4 census and certified-descent density). Every itinerary word class of depth at most four except *OOO∗* satisfies $\#\{n \le N : \mathrm{word}(n) \text{ has prefix } w\} = 2^{-|w|}N + O(N^{1-\delta_w})$ with the explicit exponents above. In particular

$$\#\{n \le N : n \text{ odd, itinerary } OOEE\} = \tfrac{N}{16} + O(N^{23/24+\varepsilon}),$$

and the class of starts with a certified descent within four steps — evens (one step), *OE* (two steps), *OOEE* (four steps) — has cardinality

$$\tfrac{N}{2} + \tfrac{N}{4} + \tfrac{N}{16} + O(N^{23/24+\varepsilon}) = \tfrac{13N}{16} + O(N^{23/24+\varepsilon}).$$

Proof. For *OOEE*, the indicator algebra gives $\#OOEE = \frac{1}{8}\sum_{n\le N \text{ odd}}(1-\psi_1)(1+\psi_2)(1+\psi_3)$ with $\psi_1 = \psi(n^{3/2})$, $\psi_2 = \psi(m^{3/2})$, $\psi_3 = \psi(v^{1/2})$; expanding gives the main term and seven sign sums bounded by Theorem 5.7. The factor $(1+\psi_2)$ vanishes precisely where J^3 would take the odd branch, so the unconditional ψ_3 is only weighted where it computes the true fourth letter. Every *OOEE* start descends within four steps by the power envelope: $3^2 < 2^4$, so Corollary 2.3 applies. The even class and the *OE* class carry the certificates of Theorem 4.1, and the three classes are disjoint. The remaining depth-≤ 4 prefixes are Theorem 5.1 (depth 1), Theorem 5.4 and Proposition 5.5 (depths 2–3), and Theorems 5.7–5.8 (depth 4). $\square$

The density 13/16 is the exact ceiling of this machinery: a word contracts iff $3^o < 2^\ell$, the method proves letters at positions 1–3 of any word plus further letters along even branches only, and the contracting minimal words with all odd letters at positions ≤ 2 are exactly E, OE, and $OOEE$, of total density $\frac{1}{2} + \frac{1}{4} + \frac{1}{16} = \frac{13}{16}$. Any further certified density requires the *OOO∗* split — a second growing layer — whose precise obstruction is isolated in Section 6.

As with Corollary 5.2, these are densities of uniform certificate classes. They are not densities of all descent certificates and not densities of starts that reach 1.

6. The remaining gap

Theorem 4.1 and Corollary 5.2 say that a uniform one- or two-step argument covers a set of density 3/4; Corollary 5.9 raises the uniformly certified class to density 13/16 at four steps. The first odd-to-odd image expands, so ordinary strong induction cannot fire on the complement. Proposition 4.4 shows that most odd-to-odd starts in a finite window still return below the start inside twenty steps. That is an observation, not Terras's theorem for J.

The depth-by-depth counting of Section 5 does assemble into a conditional Terras-style statement, and the reduction is unconditional:

Proposition 6.1 (equidistribution implies density-one descent). Let $d \ge 1$ and suppose that for every itinerary word w of length d (over all starts, first letter the parity of n),

$$\left|\#\{n \le N : \mathrm{word}_d(n) = w\} - 2^{-d}N\right| \le E_d(N).$$

Then the starts with no contracting prefix of length $\le d$ number at most

$$e^{-cd}N + 2^d E_d(N), \qquad c = 2\Big(\tfrac{\log 2}{\log 3} - \tfrac{1}{2}\Big)^2 > 0.0342.$$

Every other start $n \ge 2$ satisfies $J^t(n) < n$ for some $t \le d$ with the uniform power-envelope certificate of Corollary 2.3. Consequently, if $E_d(N) = O_d(N^{1-\delta_d})$ with $\delta_d > 0$ for every d, the set of starts admitting a finite descent certificate has natural density 1.

Proof. A word w of length d has a contracting prefix iff $3^{o_t} < 2^t$ for some $t \le d$, where o_t counts odd letters among the first t. If w has no contracting prefix then $3^{o_d} \ge 2^d$, i.e. $o_d \ge \beta d$ with $\beta = \log 2/\log 3 = 0.6309\ldots$ The number of such words is at most $2^d \Pr[\mathrm{Bin}(d, \frac{1}{2}) \ge \beta d] \le 2^d e^{-2(\beta-1/2)^2 d}$ by Hoeffding's inequality. Each word class has at most $2^{-d}N + E_d(N)$ members; summing over the non-contracting words gives the count. The density-one statement follows by letting $d \to \infty$ slowly with N (any $d(N) \to \infty$ with $2^d E_d(N)/N \to 0$). $\square$

Every E-rooted word has a contracting prefix at length one ($3^0 < 2$), so the hypothesis only ever consumes O-rooted class bounds — exactly what Section 5 proves through depth 4, except $OOO{*}$. The hypothesis beyond that is open, and its first open case has an exact shape. Differencing the $OOO{*}$ phase funnels, in every reorganization tried, into one object: for smooth c with $c \asymp kP^{9/8}$ and $c' \asymp kP^{1/8}$ on $n \sim P$,

$$K_c(P) = \sum_{\substack{n\sim P \\ n \text{ odd}}} e\big(c(n)\,\{\lfloor n^{3/2}\rfloor^{3/2}\}\big).$$

Conjecture 6.2 (kernel cancellation). $K_c(P) \ll P^{1-\delta}$ for some $\delta > 0$, uniformly over the family above. Exact-phase probes show square-root cancellation ($|K| \approx 51.9,\ 124.4,\ 1017.5$ on $5\cdot 10^3,\ 5\cdot 10^4,\ 5\cdot 10^5$ terms), but no proof is claimed.

The boundary is quantitative: the Section 5 engine reaches every itinerary letter whose phase coefficient grows slower than n (the $OE{*}{*}$ coefficient $B \asymp kn^{3/8}$ crosses integers every $\asymp P^{5/8}/k$ steps, so drift-1 intervals exist), while the $OOO{*}$ coefficient $W \asymp kn^{9/8}$ crosses integers within single steps ($W' \asymp kn^{1/8} \gg 1$) and no drift-1 interval exists. Bounding K_c is the precise remaining obstacle to the $OOO{*}$ split, hence to any certified density beyond 13/16 through this program. K_c is a bilinear correlation between the fractional parts of one nested-floor layer and a smooth weight at the scale of the next layer; we found no treatment of such an object in the nested-floor literature.

A mixed-parity heuristic, ignoring floors, gives mean log-log drift $\frac{1}{2}\log(3/4) < 0$. Finite ensembles sit near this value; hard paths are more odd-rich. This agrees qualitatively with the juggler-like random-walk model of Prasad and Prasad [12]. Fair parity is an assumption, not a dynamical theorem, and typical negative drift is not pointwise contraction.

The gap also resists the obvious finite-state attacks, and the failures are recorded, not anecdotal. Lean certifies a chain of four consecutive expanding persistent residual blocks,

$$1999 \to 5169 \to 50093 \to 193753 \to 887471$$

(`four_block_pe_1999`), so no uniform run bound of four or less exists for expanding blocks. Around such hard paths, five natural finite-state reductions fail on exact-integer counterexamples: a uniform slack tax per expanding block fails at the OOE block near $1.80\cdot 10^{32}$ with relative slack below 10^{-30} (Section 2); a compact state built from the normalized landing position θ fails because θ fills essentially all of $[0,1]$ on continuations and predicts the next parity no better than a residue class; recognition modulo 2^m fails because every residue class modulo 64 both continues and exits the iterated odd-landing sets; a 2-adic restriction from persistent-expanding history fails because the endpoint 763 has $v_2(y^3 - J(y)^2) = 1$; and the expanding word grammar is not an independent obstruction because, on $n \geq 2$, persistence already implies expansion, so that attack reduces to the tautology $J^{|w|}(n) > n$. Each failed reduction is an exact computation on named witnesses.

Two analytic routes around Conjecture 6.2 also fail, and the failures are recorded. Composing gap cells across two levels fails: on a cell where the first-level gap is constant, the second-level gap takes a new value at essentially every point (distinct-value ratio 1.0000 in exact windows), so no usable sub-cell survives. Reindexing by the image m (each m has at most one odd preimage) strips one nesting level but introduces a fiber indicator of density $\asymp m^{-1/3}$, whose sawtooth produces mode sums that must beat the sparsity exponent $1/3$ while the engine saves only $1/24$; already the first mode is fatal. Both routes are parked as negative knowledge, not retried.

The gap is therefore:

> No theorem forces every exact integer state into a contracting prefix. In particular, it is open whether almost every odd-to-odd start has a finite descent certificate. By Proposition 6.1 that would follow from all-depth parity equidistribution, whose first open case is the *OOO*$*$ kernel of Conjecture 6.2.

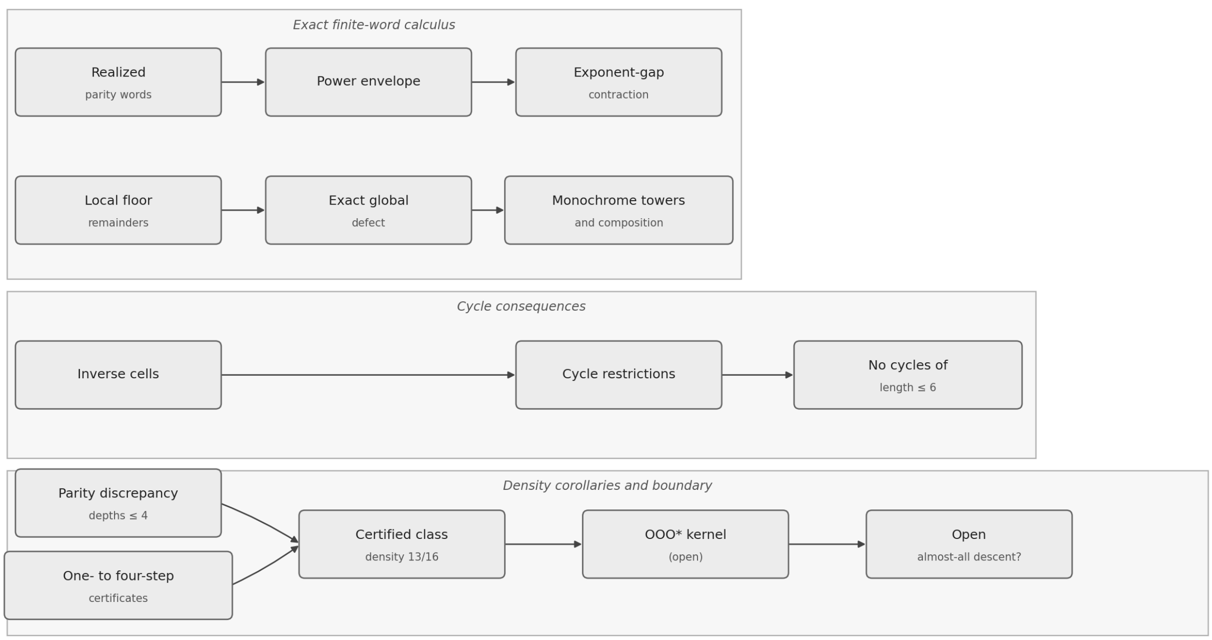

Figure 1: The theorem flow of the note. Exact finite-word identities yield contraction, rigidity, and cycle restrictions; the discrepancy calculus counts the uniform certificate classes through depth four (density 13/16) but leaves the OOO* kernel, and with it almost-all descent, open.

That is the Juggler form of the Terras question. A sharper ambient discrepancy exponent does not answer it; by Proposition 6.1 the question now rests on equidistribution at growing depth, beginning with Conjecture 6.2.

7. Software archive

Lean proofs of the exact-arithmetic theorems live at https://github.com/sneakyweasel/balanced_ternary/. The archive is not required to read the arguments above. The analytic estimates of Section 5 and Proposition 6.1 are human proofs; Lean certifies only the exact floor reductions beneath them (`GapCells.lean`) and the exact-arithmetic theorems of Sections 2–4. The scaled-integer validators for the linearization lemmas live in `src/research/juggler_sequence/two_step_parity.py` with pinned tests.

From a clone, the review object is

```
cd formal && lake build Problems.JugglerPaper
```

Acknowledgments

I used large language models extensively while drafting and revising the text, organizing companion notes, and as an interactive assistant for Lean statements, tests, and literature records. The models are not authors. Lean theorems and named computations are the certificates for those claims. The discrepancy estimates of Section 5 and Proposition 6.1 are human proofs using classical analytic inequalities; they are not Lean-certified. I take full responsibility for the contents.

References

1. C. A. Pickover, *Computers and the Imagination: Visual Adventures Beyond the Edge*, St. Martin's Press, New York, 1991, ch. 40, p. 232.
2. OEIS Foundation Inc., "Juggler sequence: if n even then $\lfloor\sqrt{n}\rfloor$ else $\lfloor n^{3/2}\rfloor$," Sequence A094683 in *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org/A094683 (accessed 28 August 2026).
3. J. C. Lagarias, "The $3x+1$ problem and its generalizations," *Amer. Math. Monthly* 92 (1985), 3–23. doi:10.1080/00029890.1985.11971528.
4. J. C. Lagarias (ed.), *The Ultimate Challenge: The $3x+1$ Problem*, American Mathematical Society, Providence, RI, 2010.
5. R. Terras, "A stopping time problem on the positive integers," *Acta Arith.* 30 (1976), 241–252. doi:10.4064/aa-30-3-241-252.
6. C. J. Everett, "Iteration of the number-theoretic function $f(2n) = n$, $f(2n+1) = 3n+2$," *Adv. Math.* 25 (1977), 42–45. doi:10.1016/0001-8708(77)90087-1.
7. T. Tao, "Almost all orbits of the Collatz map attain almost bounded values," *Forum Math. Pi* 10 (2022), e12. doi:10.1017/fmp.2022.8.
8. R. E. Crandall, "On the "$3x+1$'' problem," *Math. Comp.* 32 (1978), 1281–1292. doi:10.1090/S0025-5718-1978-0480321-3.
9. K. R. Matthews and A. M. Watts, "A generalization of Hasse's generalization of the Syracuse algorithm," *Acta Arith.* 43 (1984), 167–175. doi:10.4064/aa-43-2-167-175.
10. L. Kuipers and H. Niederreiter, *Uniform Distribution of Sequences*, Wiley-Interscience, New York, 1974.
11. H. Iwaniec and E. Kowalski, *Analytic Number Theory*, American Mathematical Society Colloquium Publications 53, Providence, RI, 2004.
12. V. Prasad and M. A. Prasad, "Estimates of the maximum excursion constant and stopping constant of juggler-like sequences," ResearchGate preprint, 2025. doi:10.13140/RG.2.2.14110.04168.

13. J. D. Vaaler, “Some extremal functions in Fourier analysis,” *Bull. Amer. Math. Soc. (N.S.)* 12 (1985), 183–216. doi:10.1090/S0273-0979-1985-15349-2.
14. S. W. Graham and G. Kolesnik, *Van der Corput's Method of Exponential Sums*, London Mathematical Society Lecture Note Series 126, Cambridge University Press, Cambridge, 1991.